WASP-50b

R-0856

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_181031` · created 2026-07-28T21:27:30+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458422.821 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.051
Transit depth (Rp/Rs)^2	3.07 +/- 1.77 [%]
Orbital Inclination (inc)	89.66 +/- 2.89
Airmass coefficient 1 (a1)	0.82 +/- 0.61
Airmass coefficient 2 (a2)	0.11 +/- 1.59
Scatter in the residuals of the lightcurve fit is	70.6 %
Best Comparison Star	#1 - [418, 142]
Optimal Aperture	0.88
Optimal Annulus	3.82
Transit Duration (day)	0.091 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.051 vs 0.1404 (deviation 0.68σ)
Duration fit vs published	0.091 d vs 0.0754 d (deviation 1.56σ)
Mid-time O–C	-4.0 min
Depth SNR	3.4
Residual scatter	70.6 %

Light curve

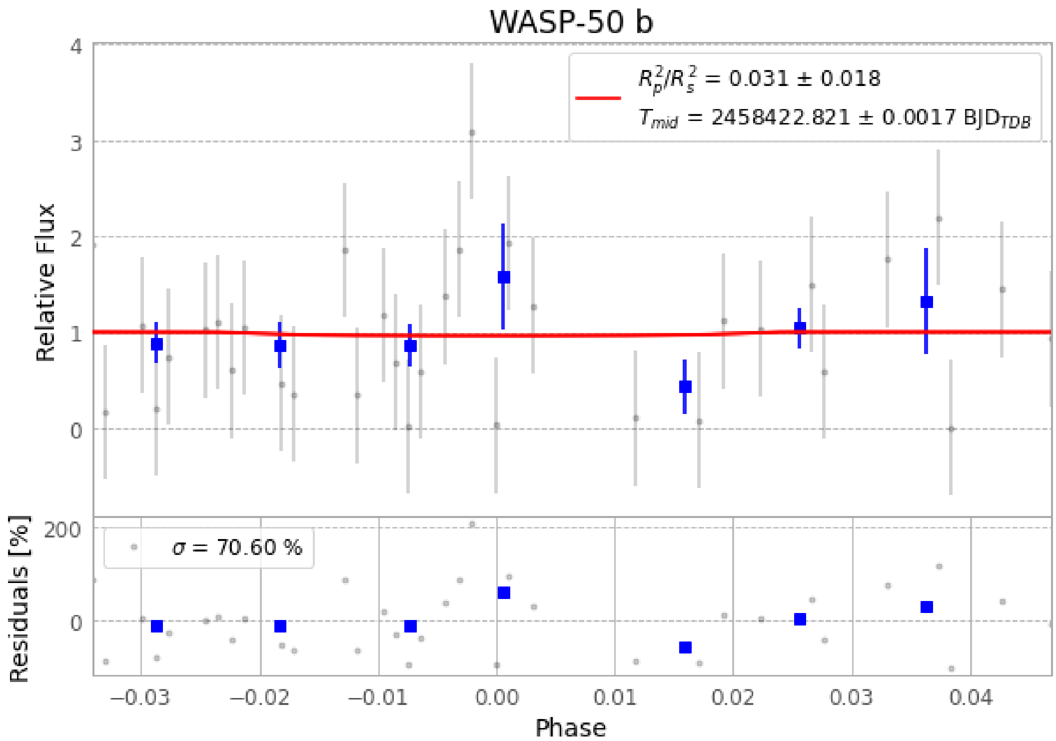

Figure 1 — FinalLightCurve_WASP-50 b_2018-10-30.png (SHA-256 e4aa31d57599cea0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [125, 384], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d83fd9fdac7f4e9c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-50181031055712.FITS` → `WASP-50181031094512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-181031.log (b164b686a672…), FinalLightCurve_WASP-50 b_2018-10-30.png (e4aa31d57599…), FinalLightCurve_WASP-50 b_2018-10-30.csv (5a2a410c00a4…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)